⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

WEEK 07

Exceptions and with-blocks

ករណីលើកលែង និង with

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This week: learn, practise, build

សប្តាហ៍នេះ រៀន អនុវត្តនិងបង្កើត

01 · Python

Exceptions and with-blocks
Understand the idea and trace examples.

02 · Practice

Practise the same Python concept.
Predict, change, diagnose and explain.

03 · Project

All-or-nothing reservation
Build, discuss and test with a peer.

One weekly component moves the final project forward.

ផ្នែកមួយរាល់សប្តាហ៍ជួយឱ្យគម្រោងចុងក្រោយរីកចម្រើន។

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Python foundations

មូលដ្ឋាន Python

Exceptions and with-blocks
Concept → trace a small example → explain its behavior

តាមដានកំហុសឆ្លងកាត់ with ហើយធានាឯកសារភាព។

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Exceptions, assignments and transaction boundaries

Exception ការកំណត់តម្លៃនិងព្រំដែន transaction

Return or raise

A return supplies a result; an exception interrupts the normal path.

Assignment order

The right side must finish before the new value is assigned.

Context behavior

A SQLite transaction can roll back database writes, not arbitrary Python changes.

Check which statements ran and which stored values changed.

ធ្វើ statement ណាបានដំណើរការ និងតម្លៃក្រោមុកណាបានផ្លាស់ប្តូរ

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Exceptions interrupt normal execution

ករណីលើកលែងផ្អាកផ្លូវធម្មតា

Signal a failure

raise stops the current path. A function does not return its normal result.

Handle at a boundary

except handles a matching failure. Assignment waits until the right side returns successfully.

Use a managed block

with asks a context manager to enter and exit a block. Cleanup behavior depends on that manager.

Understand the behavior before changing the example.

យល់ឥរិយាបថក្នុងមុនកែឧទាហរណ៍។

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Check the outcome and the state

ធ្វើលទ្ធផល និងស្ថានភាព

1

Arrange

Known starting database
Independent expectation

2

Act

Success or forced failure
Let exceptions cross the boundary

3

Inspect

Check result or error
Compare every affected table

Read the diagram, then trace the same steps in code.

អានដ្យាក្រាម រួចតាមដានជំហានដដែលក្នុងកូដ។

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Syntax to read and reuse

វាក្យសម្ពន្ធសម្រាប់អាន និងប្រើឡើងវិញ

Python / syntax	Meaning	When to use it
raise ValueError(...)	Stop the current path	Signal a known invalid value
try / except	Choose an error path	Handle a specific exception
with db:	A transaction boundary	Observe normal or failed exit
assertRaises(...)	Check an expected error	Also inspect the resulting state

Python Basics · sections 09, 13, 17 · use only when you need a lookup.

Python Basics ផ្នែក 09, 13, 17៖ មើលយោងតែពេលចាំបាច់។

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A failed call leaves its assignment unfinished

ការហៅដែលបរាជ័យមិនបញ្ចប់ការកំណត់តម្លៃ

```
01 def checked_count(value):
02     if value < 0:
03         raise ValueError("negative count")
04     return value
05 count = 5
06 try:
07     count = checked_count(-1)
08 except ValueError:
09     print("rejected")
10 print(count)
```

1. Expected: rejected, then 5.
2. The right side raises before count is reassigned.
3. Change -1 to 2 and predict the new output.

Predict → execute → explain the value at each step.

ព្យាករ → ដំណើរការ → ពន្យល់តម្លៃនៅជំហានមួយៗ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

assertRaises: check the exception and state

assertRaises៖ ធ្វើការណាំពីលើកលែង និងស្ថានភាព

```
01  import unittest
02
03  def reserve_count(available, requested):
04      if requested > available:
05          raise ValueError("not enough containers")
06      return available - requested
07
08  check = unittest.TestCase()
09  available = 3
10  with check.assertRaises(ValueError):
11      available = reserve_count(available, 4)
12  check.assertEqual(available, 3)
```

1. assertRaises checks the exception type.
2. Failed assignment keeps the old value.
3. Database rollback needs its own state check.

Check the expected exception and the state left afterward.

ធ្វើការណាំពីលើកលែងដឹង និងស្ថានភាពក្រោយកំហុស។

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Hands-on practice

លំហាត់អនុវត្តផ្ទាល់

Week_07.ipynb · Session 2
Predict → run → change → explain

ព្យាករ រត់ ប្តូរ និងពន្យល់លទ្ធផល។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Your turn: practise the same Python skills

ដល់វេនអ្នក អនុវត្តជំនាញ Python ដដែល

P1–P2 · Easy

Use the approach hints. Write the code yourself in a blank cell.

P3 · Core build

Implement a complete small program. Write expected results and assertions.

P4–P5 · Optional

Apply the skill to a new case, then attempt a harder challenge if time permits.

Write your own code, predict the result, then test and explain it.

សរសេរកូដផ្ទាល់ ព្យាករលទ្ធផល ឆ្លុះសាក និងពន្យល់។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Check your work before asking for help

ធ្វើការងារមុនសុំជំនួយ

A complete implementation

Write the whole operation. Do not stop at one changed character.

Independent expected results

Calculate a normal case and an edge case before running your program.

A peer can explain it

Swap driver and reviewer. Explain one line and one test.

Trace raise, except and a with-block boundary; inspect the remaining state.

តាមដាន raise, except និងព្រំដែន with; ធ្វើស្ថានភាពនៅសល់។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Project studio

វគ្គសិក្សាគម្រោង

All-or-nothing reservation
Short design briefing → most time for your work

កក់ទាំងសំណើឬទុកឱ្យទាំងអស់ទៅឱ្យប្រាំ

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The people’s task this week

កិច្ចការរបស់អ្នកប្រើនាសប្តាហ៍នេះ

1

Someone needs help

A booking must not leave only part of a request set aside.

2

This week, you will

Accept the whole booking, or leave the records unchanged if it cannot finish.

3

A useful result

Another volunteer sees consistent available counts.

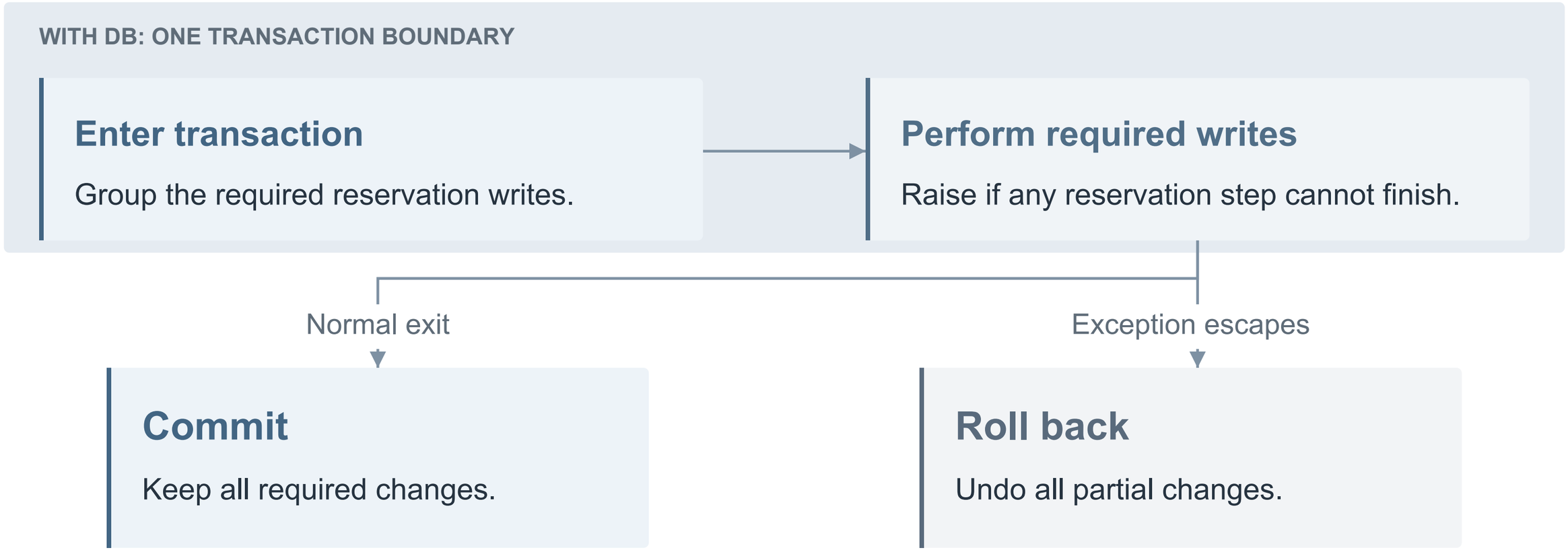
Use the supplied starting files. Focus on one useful result and check it.

ប្រើកសារចាប់ផ្តើមដែលផ្តល់។ ផ្តោតលើលទ្ធផលមានប្រយោជន៍មួយ ហើយធ្វើតេស្ត។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Let a required failure escape before handling it

ឱ្យកំហុសចេញពីប្រតិបត្តិការមុនដោះស្រាយ



Catch the failure after the transaction has had a chance to roll back.

ចាប់កំហុសក្រោយប្រតិបត្តិការមានឱកាស rollback។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Reference: Observe rollback after a deliberate interruption

ឯកសារយោង៖ ឃើញ rollback ក្រោយបង្កាច់ហ៊ុសសាកល្បង

```python
import sqlite3
db = sqlite3.connect(":memory:")
db.execute("CREATE TABLE availability(containers INT)")
with db: db.execute("INSERT INTO availability VALUES (5)")
try:
    with db:
        db.execute("UPDATE availability SET containers=3")
        raise RuntimeError("practice interruption")
except RuntimeError: pass
row = db.execute("SELECT * FROM availability").fetchone()
assert row == (5,)
db.close()
```

1. Commit the opening state first.
2. Let failure escape the with block.
3. Read the state after handling the error.

An escaping exception rolls back writes in the transaction.

ហ៊ុសដែលចេញផ្លូវធ្វើ rollback ការសរសេរក្នុងប្រតិបត្តិការ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Conditional UPDATE with placeholders

UPDATE តាមលក្ខខណ្ឌជាមួយកន្លែងដាក់តម្លៃ

```
UPDATE offer_batches
SET available_containers = available_containers - ?
WHERE batch_id = ?
  AND available_containers >= ?;
```

1. Bind requested count, batch ID, requested count
2. Inspect cursor.rowcount after execute
3. Zero does not automatically raise a failure

A zero-row update must be inspected and handled explicitly.

UPDATE បានសង្ស័យជួរដែលរងផល និងដោះស្រាយដោយច្បាស់។

rowcount: one row changed

rowcount៖ ជួរមួយបានផ្លាស់ប្តូរ

```
import sqlite3
db = sqlite3.connect(":memory:")
db.execute("CREATE TABLE offers(id TEXT, available INT)")
db.execute("INSERT INTO offers VALUES ('DEMO',5)")
cur = db.execute(
    "UPDATE offers SET available=available-? "
    "WHERE id=? AND available>=?", (2, "DEMO", 2))
assert cur.rowcount == 1
row = db.execute("SELECT * FROM offers").fetchone()
assert row[-1] == 3
db.close()
```

1. Inspect the returned cursor.
2. rowcount is 1 after a match.
3. Query again to observe the value.

A matching conditional UPDATE changes one row.

UPDATE តាមលក្ខខណ្ឌដែលត្រូវ ផ្លាស់ប្តូរជួរ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

rowcount: no row matched

rowcount៖ គ្មានជួរត្រូវលុបឡើយ

```
01 import sqlite3
02 db = sqlite3.connect(":memory:")
03 db.execute("CREATE TABLE offers(available INT)")
04 db.execute("INSERT INTO offers VALUES (3)")
05 cur = db.execute(
06     "UPDATE offers SET available=available-? "
07     "WHERE available>=?", (4, 4))
08 assert cur.rowcount == 0
09 row = db.execute("SELECT * FROM offers").fetchone()
10 assert row[-1] == 3
11 db.close()
```

1. rowcount is 0 when the condition fails.
2. The stored value stays the same.
3. The caller decides whether to raise.

No matching row means no update; it does not raise automatically.

គ្មានជួរត្រូវ៖ គ្មានការកែ; មិន raise ដោយស្វ័យប្រវត្តិ។

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Your implementation TODO

ការងារដែលអ្នកត្រូវសរសេរក្នុង

Supplied starting point

Database fixture starts Q11=4, Q12=4; schema, starter and failure-injection hook are supplied.

You implement / do

Validate review, counts and batches; apply guarded updates in one transaction. Raise on shortage; let failure escape the with-block before handling it.

Check the evidence

Unreviewed, shortage and injected failure must leave state unchanged. Success Q11:2 + Q12:1 → remaining 2,3; one header and two lines.

Also check zero counts, unknown batches and repeated batch lines.

ពិនិត្យផងដែរសូន្យ ទូរទិស្សន៍ និងជួរទូរទិស្សន៍ដែលជាន់។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Show the component working

បង្ហាញថាផ្នែកនេះដំណើរការ

Step	What to show
1 · Locate	Identify the input, this component and its next user.
2 · Implement	Implement reserve in one transaction; reject bad requests.
3 · Check	Success leaves 2/3; refusals leave every table unchanged.
4 · Explain	Show your code, its actual result and why it behaves that way.

Also check zero counts, unknown batches and repeated batch lines.

ធ្វើចំនួនសូន្យ ទូស្តើស្គាល់ និងជួទូដែលផង។

Further reading in English

ឯកសារអានបន្ថែមជាភាសាអង់គ្លេស

1 SQLite transactions

https://www.sqlite.org/lang_transaction.html

2 Python sqlite3 connection context manager

https://docs.python.org/3/library/sqlite3.html#how-to-use-the-connection-context-manager

Read one relevant section, then bring an example and a question.

អានផ្នែកដែលពាក់ព័ន្ធមួយ យកឧទាហរណ៍មួយ និងសំណួរមួយមកវិញ។